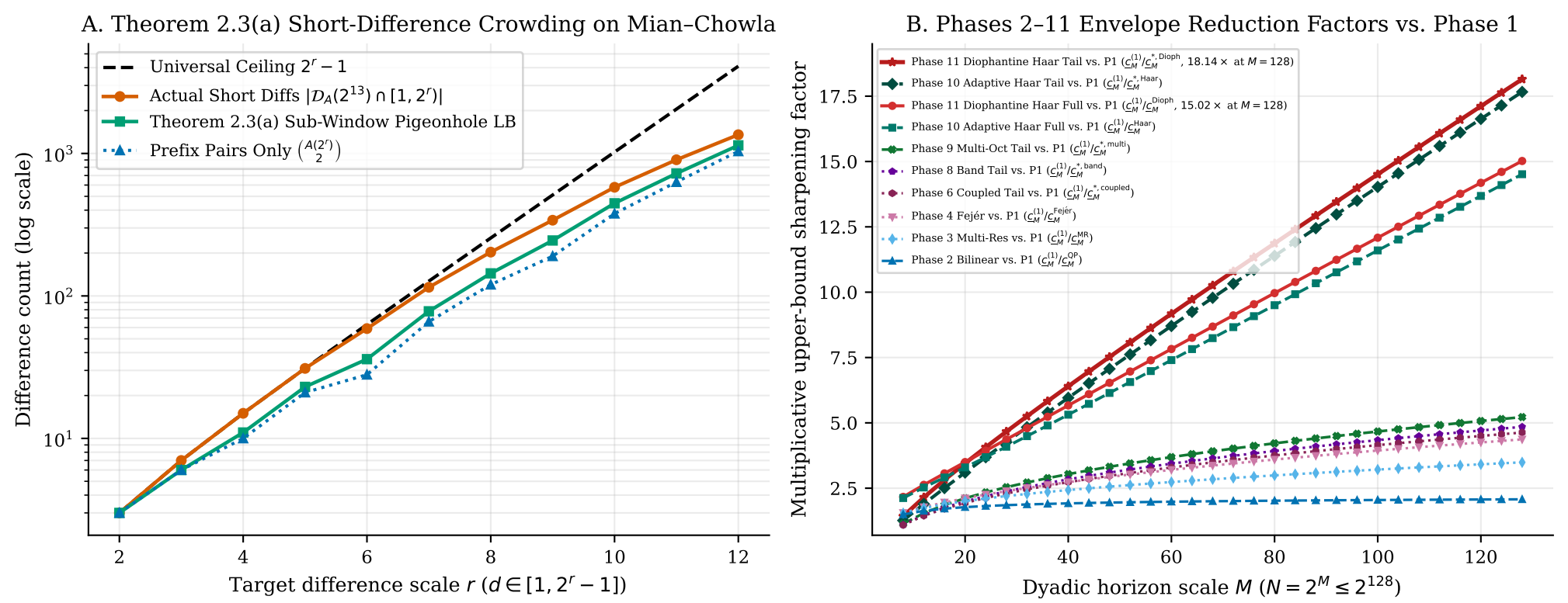

A. Theorem 2.3(a) Short-Difference Crowding on Mian–Chowla
Universal Ceiling $2^r - 1$
Actual Short Diffs $|\mathcal{D}_A(2^{13}) \cap [1, 2^r)|$
Theorem 2.3(a) Sub-Window Pigeonhole LB
Prefix Pairs Only $\binom{A(2^r)}{2}$
Difference count (log scale)
10^3
10^2
10^1
2
4
6
8
10
12
Target difference scale r ($d \in [1, 2^r - 1]$)
B. Phases 2–11 Envelope Reduction Factors vs. Phase 1
Phase 11 Diophantine Haar Tail vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{*,\mathrm{Dioph}}$, 18.14× at $M = 128$)
Phase 10 Adaptive Haar Tail vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{*,\mathrm{Haar}}$)
Phase 11 Diophantine Haar Full vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{\mathrm{Dioph}}$, 15.02× at $M = 128$)
Phase 10 Adaptive Haar Full vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{\mathrm{Haar}}$)
Phase 9 Multi-Oct Tail vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{*,\mathrm{multi}}$)
Phase 8 Band Tail vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{*,\mathrm{band}}$)
Phase 6 Coupled Tail vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{*,\mathrm{coupled}}$)
Phase 4 Fejér vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{\mathrm{Fejér}}$)
Phase 3 Multi-Res vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{\mathrm{MR}}$)
Phase 2 Bilinear vs. P1 ($\underline{c}_M^{(1)}/\underline{c}_M^{\mathrm{QP}}$)
Multiplicative upper-bound sharpening factor
17.5
15.0
12.5
10.0
7.5
5.0
2.5
20
40
60
80
100
120
Dyadic horizon scale M ($N = 2^M \le 2^{128}$)